

ECE 635 Homework 7 Answers

2025 Fall

1. Answer is given in the homework problem

2. " " " " " " "

3. For the source use cylindrical primed coordinates
For the far field use spherical unprimed coordinates

$$A_z = \frac{\mu}{4\pi} \int_{z'=-l/2}^{l/2} \int_{\phi'=0}^{2\pi} \int_{\rho'=0}^{\infty} J_s \delta(\rho-a) \hat{z} e^{-jk[\rho^2+\rho'^2-2\rho\rho'\cos(\phi-\phi')+(z-z')^2]^{1/2}} \rho d\rho d\phi' dz'$$

$$\hat{n} \times (\vec{E}^{\text{ext}} - \vec{E}^{\text{int}}) \Big|_{\rho=a, -l/2 \leq z \leq l/2} = 0 \quad \text{from interior-exterior cases}$$

$$\hat{\rho} \times [\vec{E}^{\text{inc}} + \vec{E}^{\text{scat}}] \Big|_{\rho=a, -l/2 \leq z \leq l/2} = 0$$

$$\hat{\rho} \times \vec{E}^{\text{inc}} = -\vec{E}^{\text{scat}} \hat{\rho} \times \hat{z}$$

$$= -E_z \hat{\phi}$$

$$E_z' = \frac{j\omega}{R^2} (R^2 A_z + \frac{\partial}{\partial z} A_z), \quad \text{after } \rho' \text{ integration} \quad |\vec{r} - \vec{r}'| = [2a^2 - 2a^2 \cos \phi' + z - z']^{1/2}$$